Descriptive Statistics

Assignment

Q1: Understanding Central Tendency
A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12]. What is the most representative value of their weekly sales, and why?

Ans; Mean= $\frac{10+12+11+15+14+13+12}{7} = \frac{87}{7} =$ **12.43**

Median= Arranging data in ascending order:
(10,11,12,12,13,14,15)The middle value (4th term) = **12**

Mode= The number that appears most often = **12**

The most representative value of weekly sales is 12 dozen muffins because the mean, median, and mode are all close to 12, showing that sales are consistent around this value. Therefore, 12 dozen best represent the bakery's daily sales.

Q2: Mean in Real Life
A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19]. What is the mean score, and what does it tell us about the class's performance?

Mean= $\frac{12+15+14+16+18+20+19}{7} = \frac{114}{7} =$ **16.29**

The mean score is 16.29 marks. It is the average performance of the class.
Most students scored **around 16 marks**, meaning the class performed consistently.
It helps the teacher understand the **overall level of understanding** — not just individual highs or lows.

Q3: Mode in Real Life
A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9]. What is the mode, and why is this information useful for the store manager ?

The **mode** is the number that appears most frequently, in other words the frequency.
Here,[7,7,8,8,8,9,9,10]
8 occurs **three times**, more than any other size. So, the **mode = 8**.

The mode shows the **most frequent value** in a dataset. In real life, this helps the store manager understand which shoe size is **most in demand**. Knowing this allows them to **stock in more of size 8**, ensuring better inventory management and customer satisfaction.

Q4: Median in Real Life
A car dealer notes the prices of used cars: [$8,000, $9,500, $10,200, $11,000, $50,000]. Why is the median a better measure than the mean in this case? Calculate the median.
The **median** is the middle value.Here, the third value = **$10,200**.

The **median price** is **$10,200**.

The **mean** would be pulled upward by the extremely high price of **$50,000**,as it is the outlier giving a misleading average.The **median**, however, is **not affected by outliers** — it represents the **true central value** of the data.In this case, the median gives a **more realistic value** of typical car prices, while the mean would exaggerate the average due to one unusually expensive car.

Q5: Dispersion Introduction
A student times how long it takes to finish a puzzle each day: [25, 30, 27, 35, 40]. What does the range tell us about the variation in the student's puzzle-solving time?

Range= Highest value−Lowest value= 40−25= 15
The range of puzzle-solving time is 15 minutes.

The range shows the **spread or dispersion** of data , how much the values vary from each other. In this case, the student's puzzle-solving time varies by **15 minutes** between the fastest and slowest day. This means the student's performance is **not perfectly consistent**, and there's noticeable variation in how long it takes to complete the puzzle each day.

Q6: Range in Action
A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120]. Find the range. How can this help the farmer in planning his packaging?

Range= Highest value−Lowest value= 120−98= 22
The **range** of apple harvest weights is **22 kg**.

The range shows the **extent of variation** in the data — how much the values differ from each other. In real life, this helps the farmer understand the **fluctuation in weekly harvests**. A range of 22 kg means the harvest quantity changes noticeably from week to week. This insight helps the farmer to **plan storage, packaging, and sales** more efficiently by anticipating weeks of higher or lower yield.

Q7: Variance for Decision-Making
Two delivery companies track

delivery delays (in minutes). Company A: variance = 6 Company B: variance = 15 Which company is more consistent, and why?

Company A is more consistent.

variance measures how much the data points differ from the mean — it shows the **degree of variability**. How other data are varying with respect to your mean/average.
A **lower variance** means the data values are closer to the average, indicating **greater consistency** and **predictability**.

Since Company A's variance (6) is smaller than Company B's (15), Company A's delivery times vary less, making it **more reliable and stable in performance.**

Q8: Standard Deviation in Context

A finance student compares the daily price fluctuations of two cryptocurrencies

Coin A: standard deviation = $30

Coin B: standard deviation = $120

Which coin is riskier to invest in, and why?

Standard deviation measures how much the daily prices fluctuate around the average.

A higher standard deviation means greater variability ,prices move more unpredictably.

Coin B is **riskier to invest in** because its standard deviation ($120) is much higher than Coin A's ($30).

This indicates Coin B's price changes are more volatile, meaning it has more potential for both higher gains and higher risk.

In finance, investors often prefer assets with lower standard deviation when they want stability.
A higher standard deviation suits risk-takers seeking potentially higher returns.

Q9: Combining Measures

A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410].

Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Mean= $\frac{400+420+390+450+410}{5} = \frac{2070}{5} = 414$

$\sigma = \sqrt{}((400 - 414)^2 + (420 - 414)^2 + (390 - 414)^2 + (450 - 414)^2 + (410 - 414)^2) / 5$

$\sigma = \sqrt{}(((- 14)^\wedge 2 + 6^\wedge 2 + (- 24)^\wedge 2 + 36^\wedge 2 + (- 4)^\wedge 2) / 5)$

$\sigma = \sqrt{}(2120 / 5) = \sqrt{}(424) = 20.6$

So, the **standard deviation ≈ 20.6 kWh**.

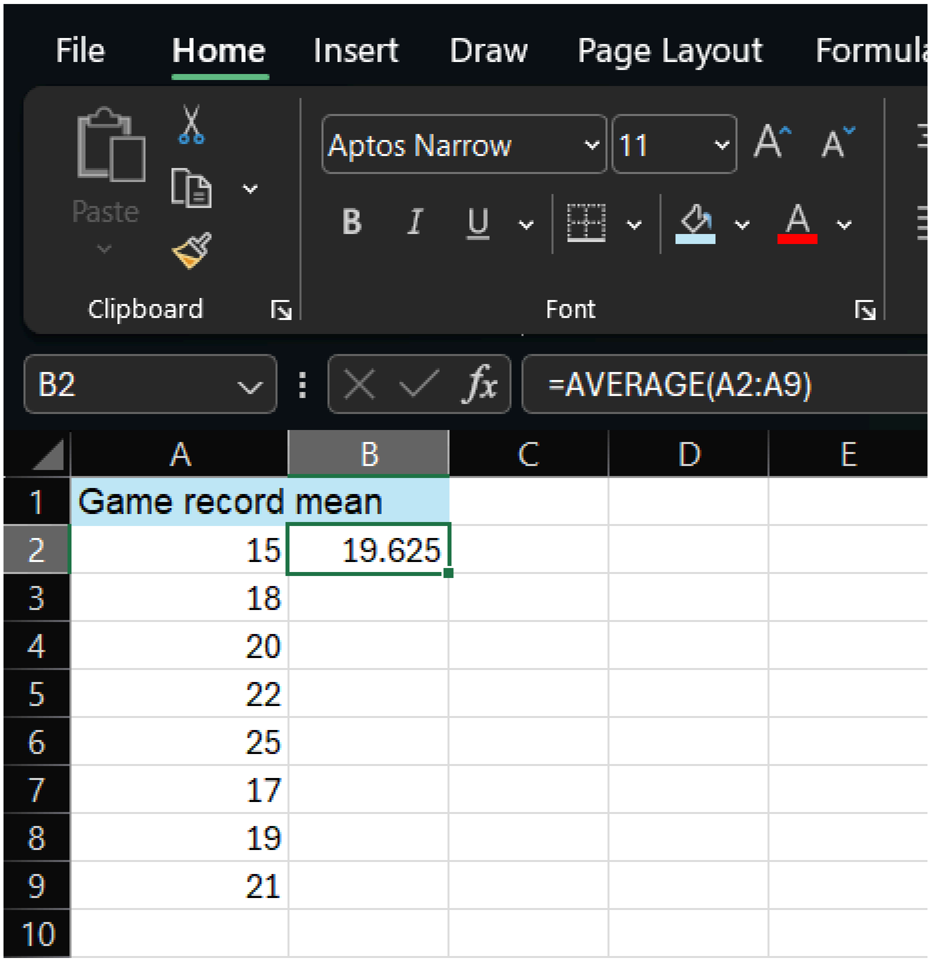
- The mean 414 kWh shows the family's **average monthly consumption**.
- The standard deviation (≈ 20.6 kWh) indicates **low variation**, meaning their electricity use is **fairly consistent** month to month.
- This suggests a **stable energy pattern**, with no major spikes or drops in usage.

Q10: Practical Application

A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21].

Find the mean, median, mode, range, and standard deviation. What insights can these measures provide about the player's scoring performance

The player's average score is **19.6 points per game**, with a **standard deviation of ≈ 2.9 points**, showing only minor fluctuations. The **range of 10 points** indicates moderate variation between the highest and lowest scores. Since there is **no mode**,because of no repetition of any value, the player's performance varies slightly each game but remains consistent overall. These measures suggest the player is **steady and reliable**, maintaining a stable scoring pattern across matches.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2 15 19.625
3 18
4 20
5 22
6 25
7 17
8 19
9 21
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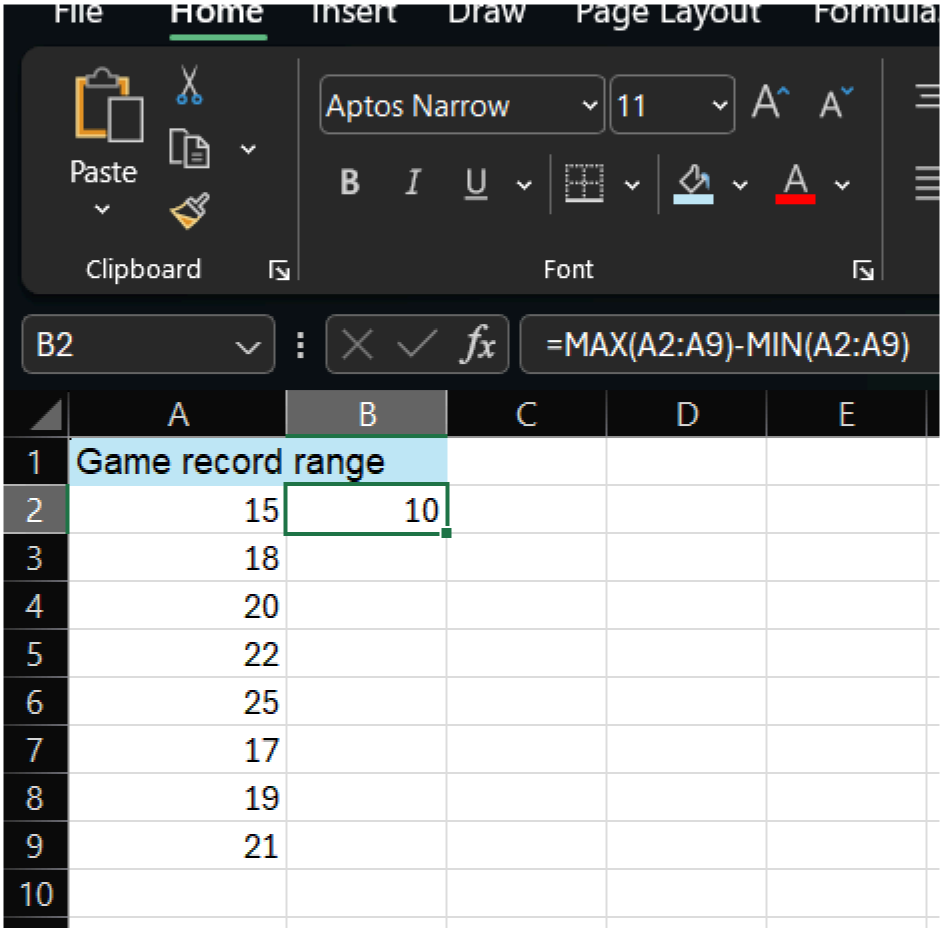
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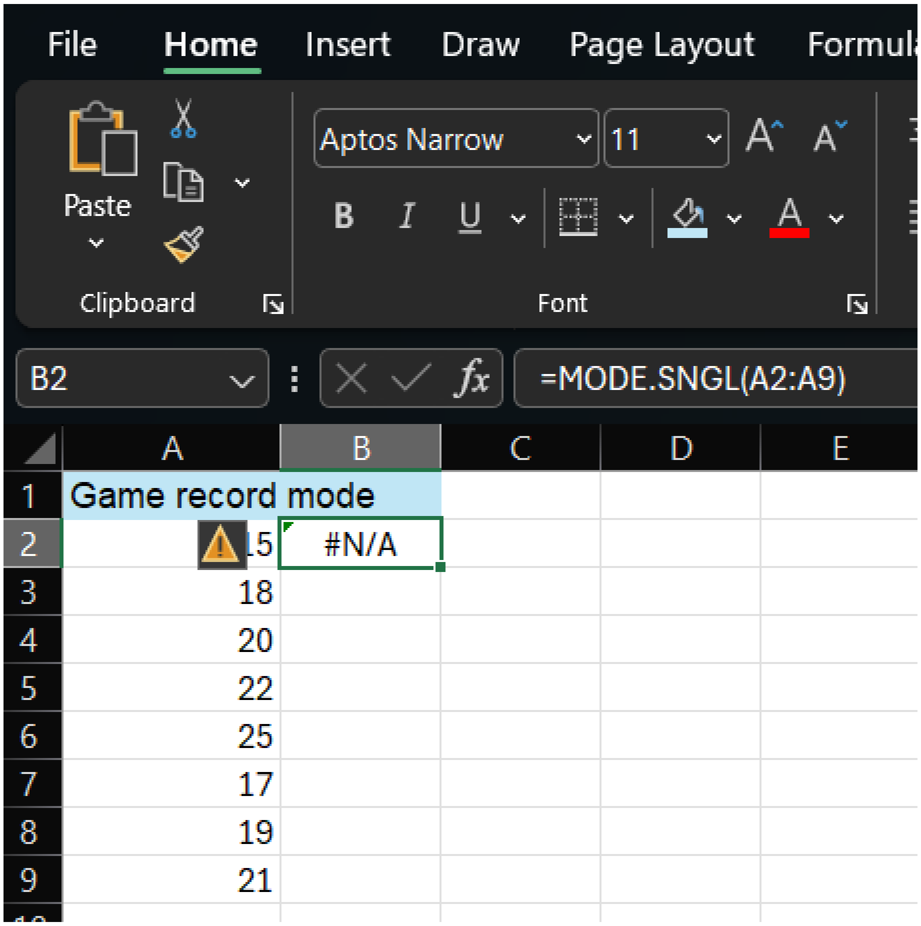
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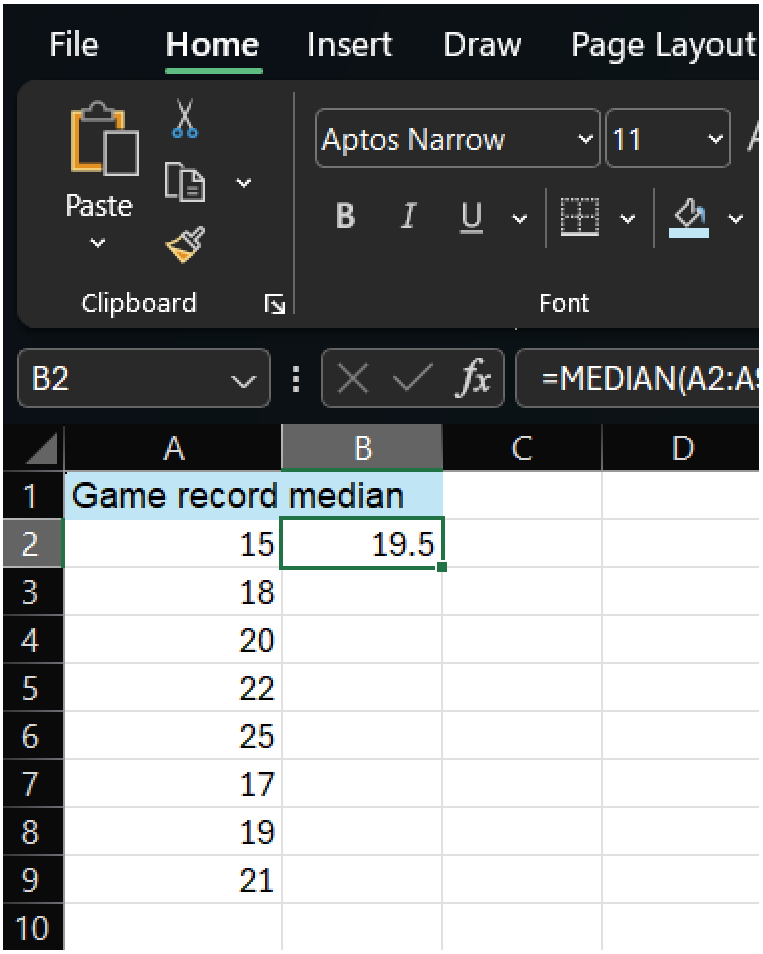
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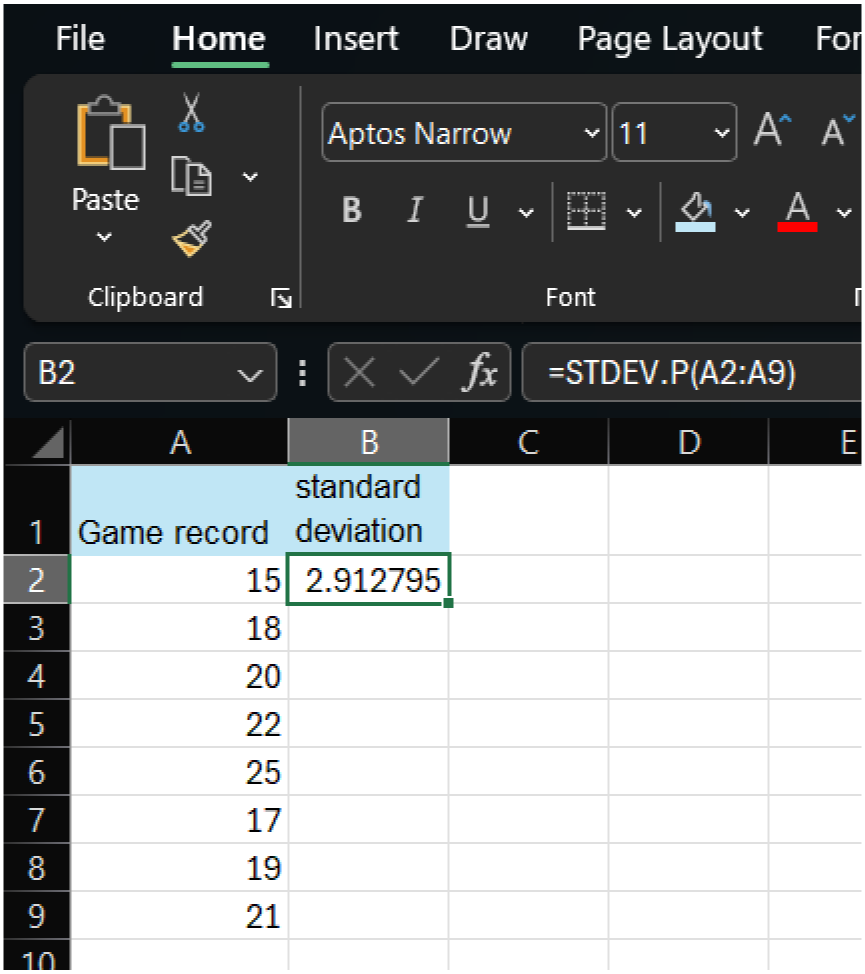
1 Game record median
2 15 19.5
3 18
4 20
5 22
6 25
7 17
8 19
9 21
10



	A	B	C	D
1	Game record	standard deviation		
2	15	2.912795		
3	18			
4	20			
5	22			
6	25			
7	17			
8	19			
9	21			